

Cloud Computing Tutorial 3

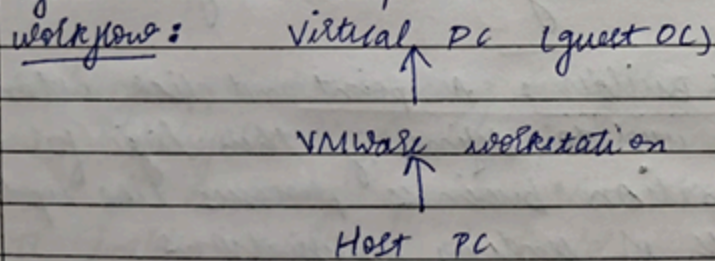
Create a Virtual Machine Using VMware Workstation Pro 12
and Install Python

Observation:

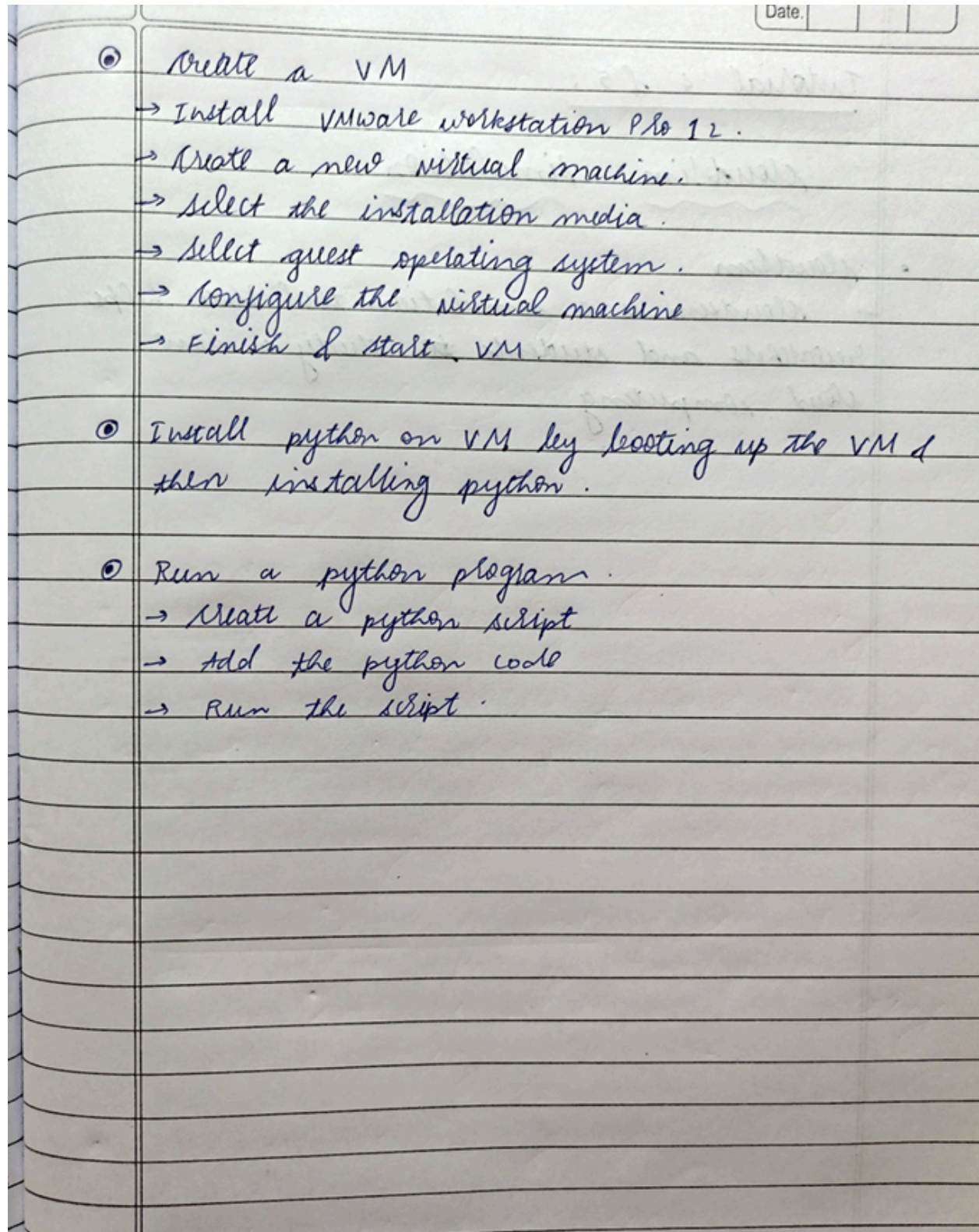
Tutorial 3:

creating a Virtual Machine (VM) using VMware Workstation Pro

- What is Virtual Machine (VM)?
 - A virtual machine is basically a computer inside another computer.
 - It behaves like a real computer because it:
 - * has its own OS (Windows, Ubuntu, etc.)
 - * has its own RAM
 - * has its own storage
 - * has its own & can install software.
- VMware Workstation Pro is a software that lets you create and run virtual machines easily on your PC. It acts like a manager for your virtual computers.



- Benefits of using VMware Workstation Pro.
 - Run multiple operating systems at the same time.
 - Doesn't require dual boot.
 - Offers safe testing environment.
 - Save current state to restore later as snapshots.
 - Better performance of Workstation Pro and supports heavy workload.



Oracle Virtual Machine Demonstration:

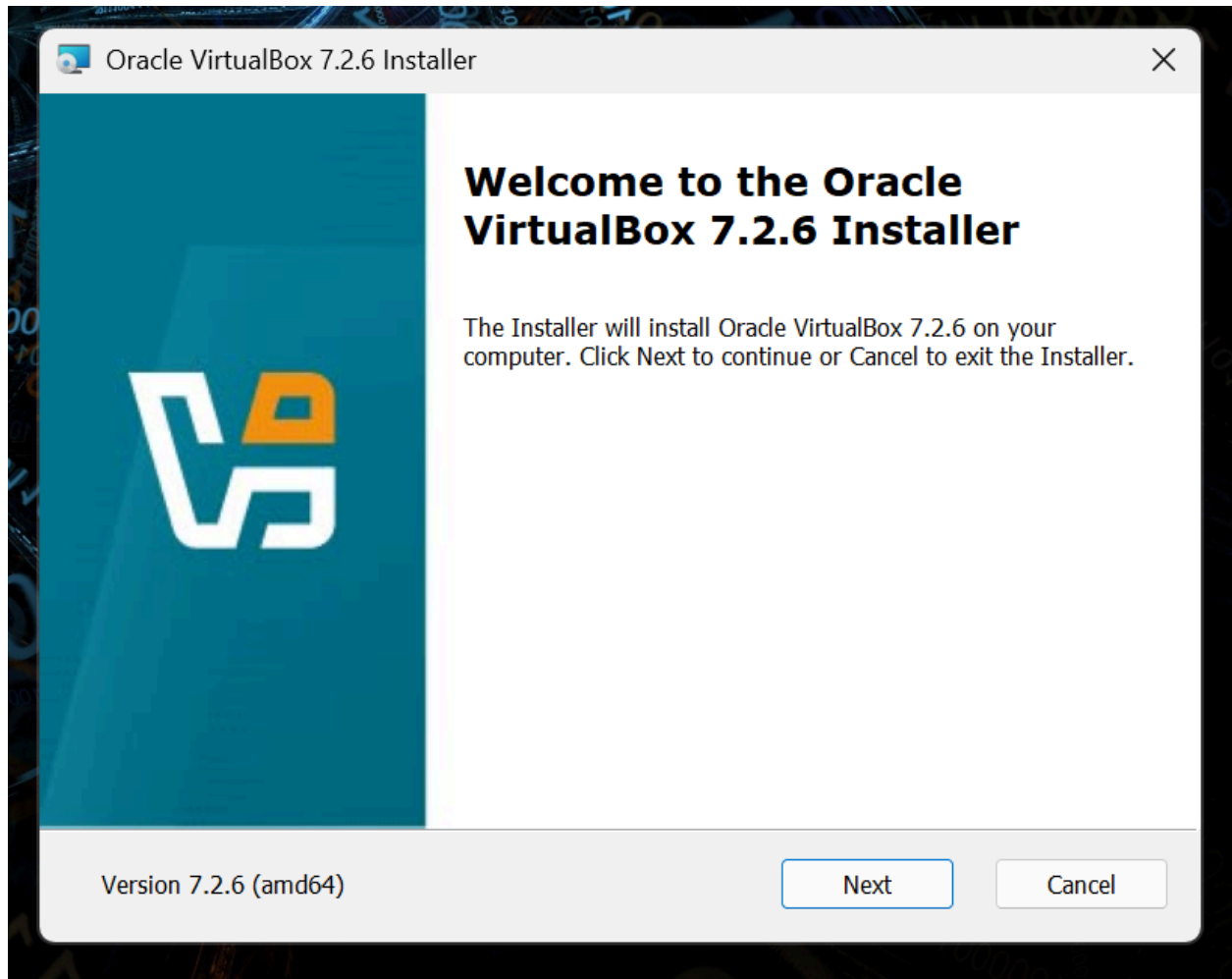


Fig 1.1 Oracle VirtualBox Installation

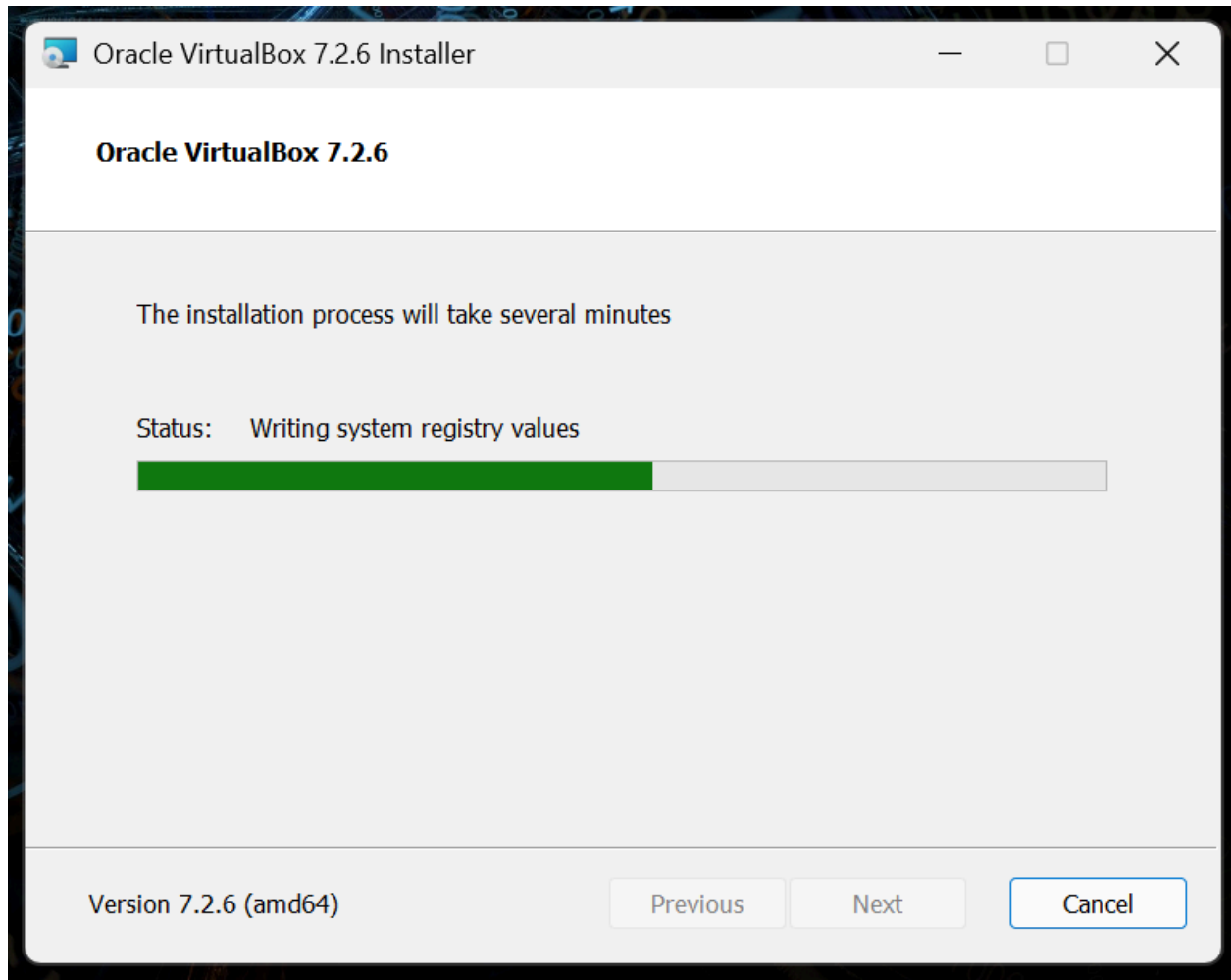


Fig 1.2 Installing Oracle VirtualBox

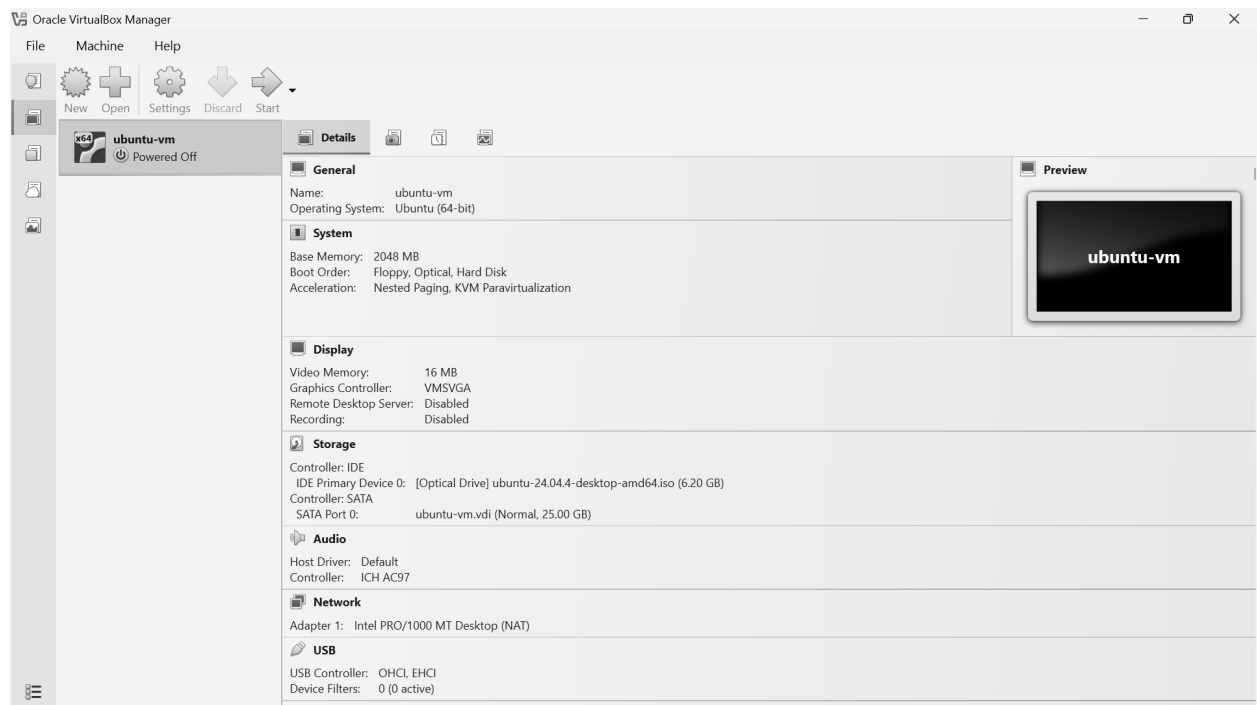
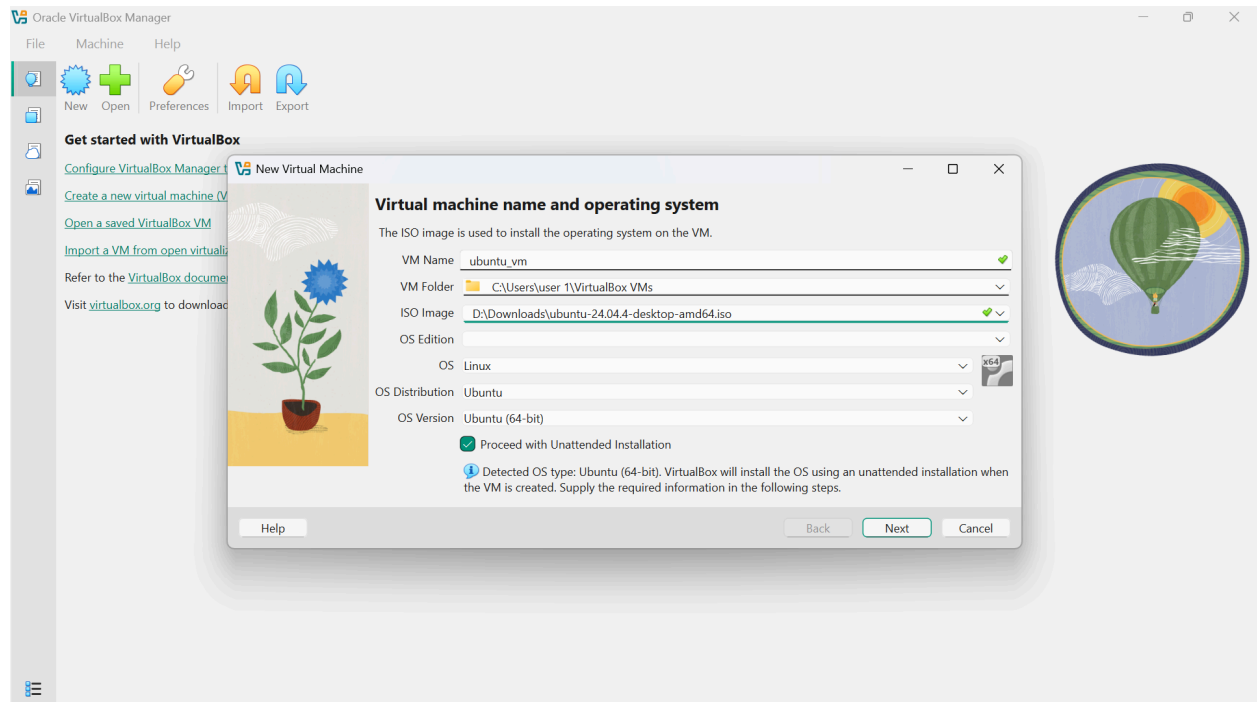


Fig 1.2 Turning on Ubuntu virtual machine on Oracle VirtualBox Manager

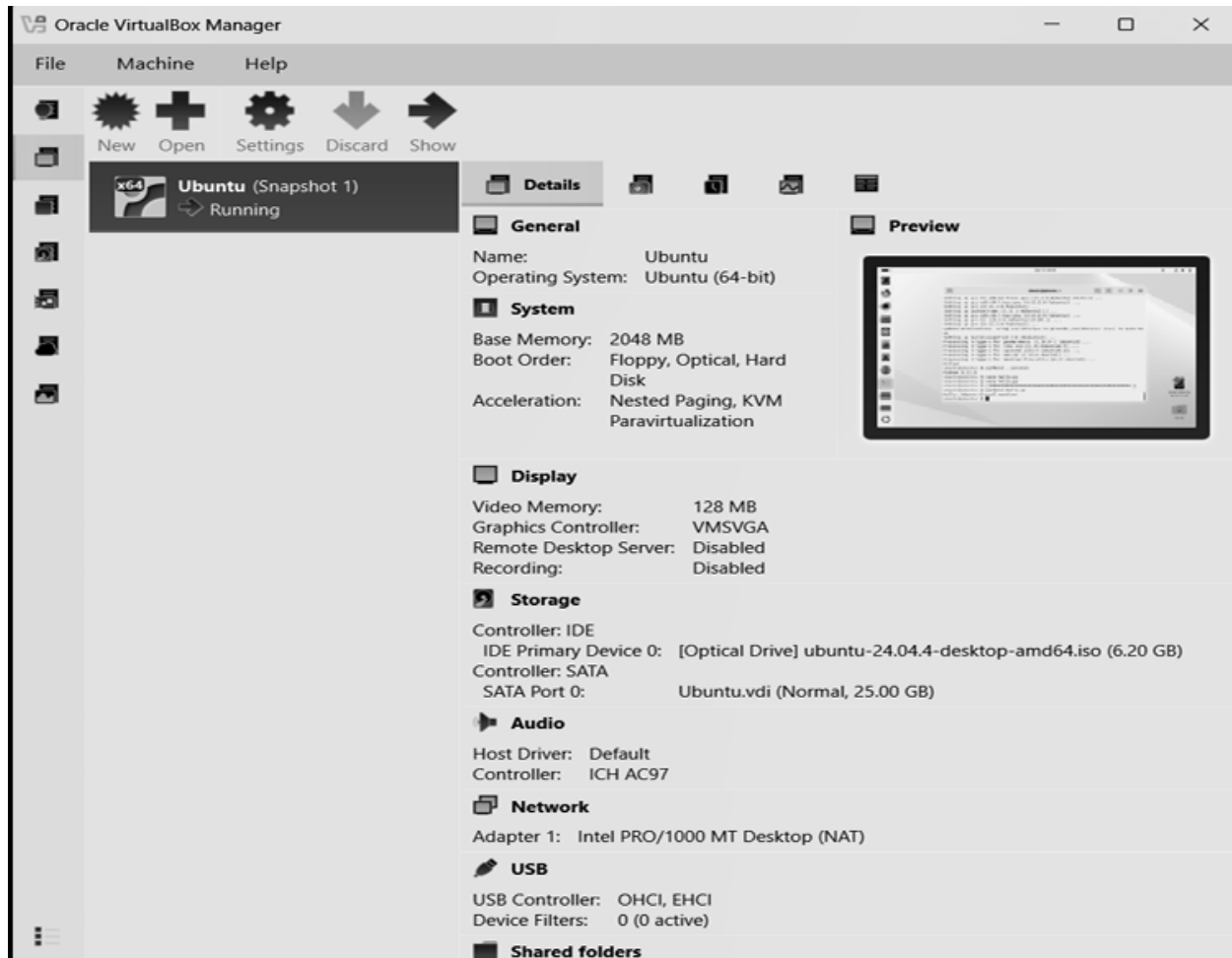


Fig 1.3 Ubuntu virtual machine running

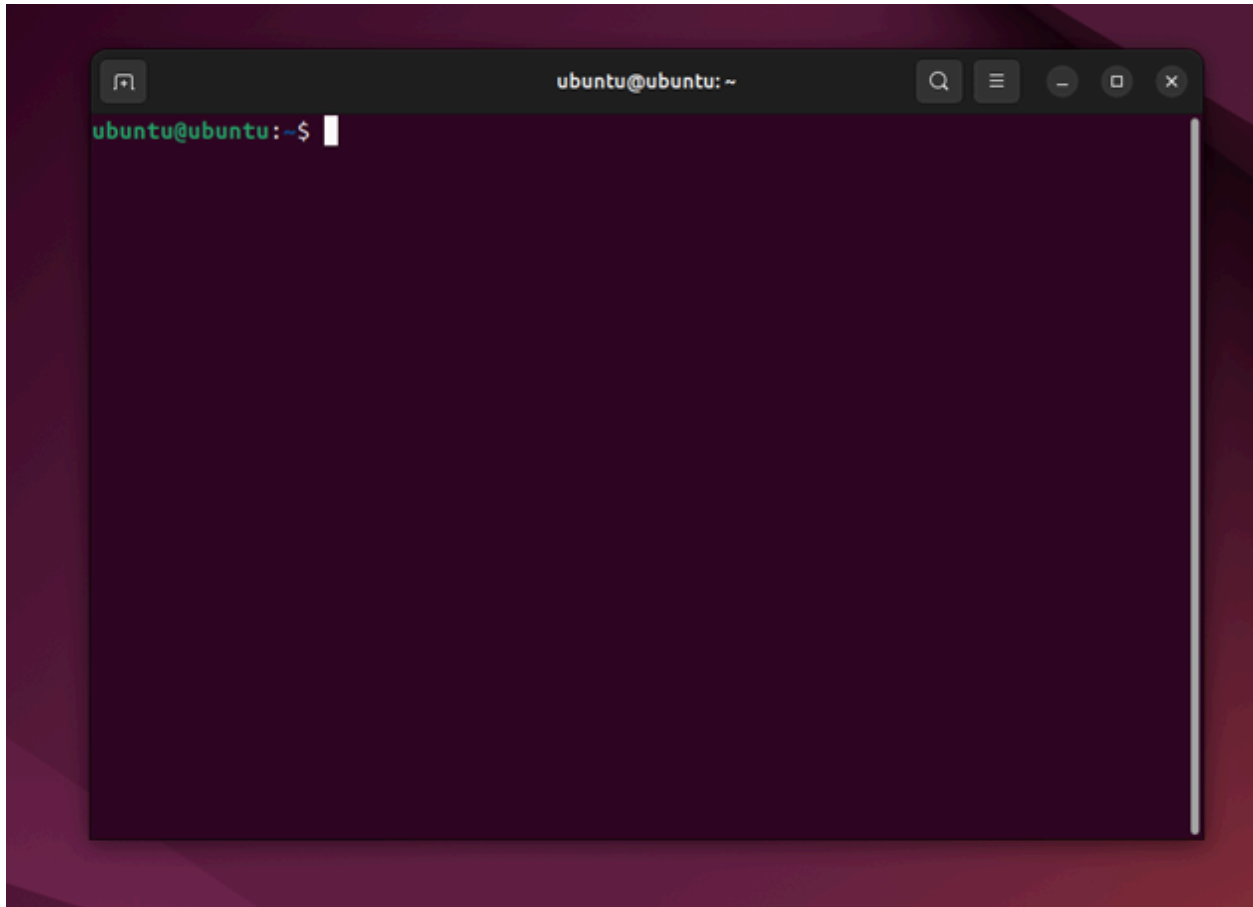


Fig 1.4 Ubuntu Terminal

```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ sudo apt update  
Ign:1 cdrom://Ubuntu 24.04.4 LTS _Noble Numbat_ - Release amd64 (20260210) noble  
InRelease  
Hit:2 cdrom://Ubuntu 24.04.4 LTS _Noble Numbat_ - Release amd64 (20260210) noble  
Release  
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]  
Get:5 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1579  
kB]  
Get:6 http://security.ubuntu.com/ubuntu noble-security/main i386 Packages [400 k  
B]  
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [253  
kB]  
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21  
.5 kB]  
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata  
[10.7 kB]  
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe i386 Packages [7  
91 kB]  
Hit:11 http://archive.ubuntu.com/ubuntu noble InRelease  
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages  
[1170 kB]  
Get:13 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]  
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en  
[225 kB]
```

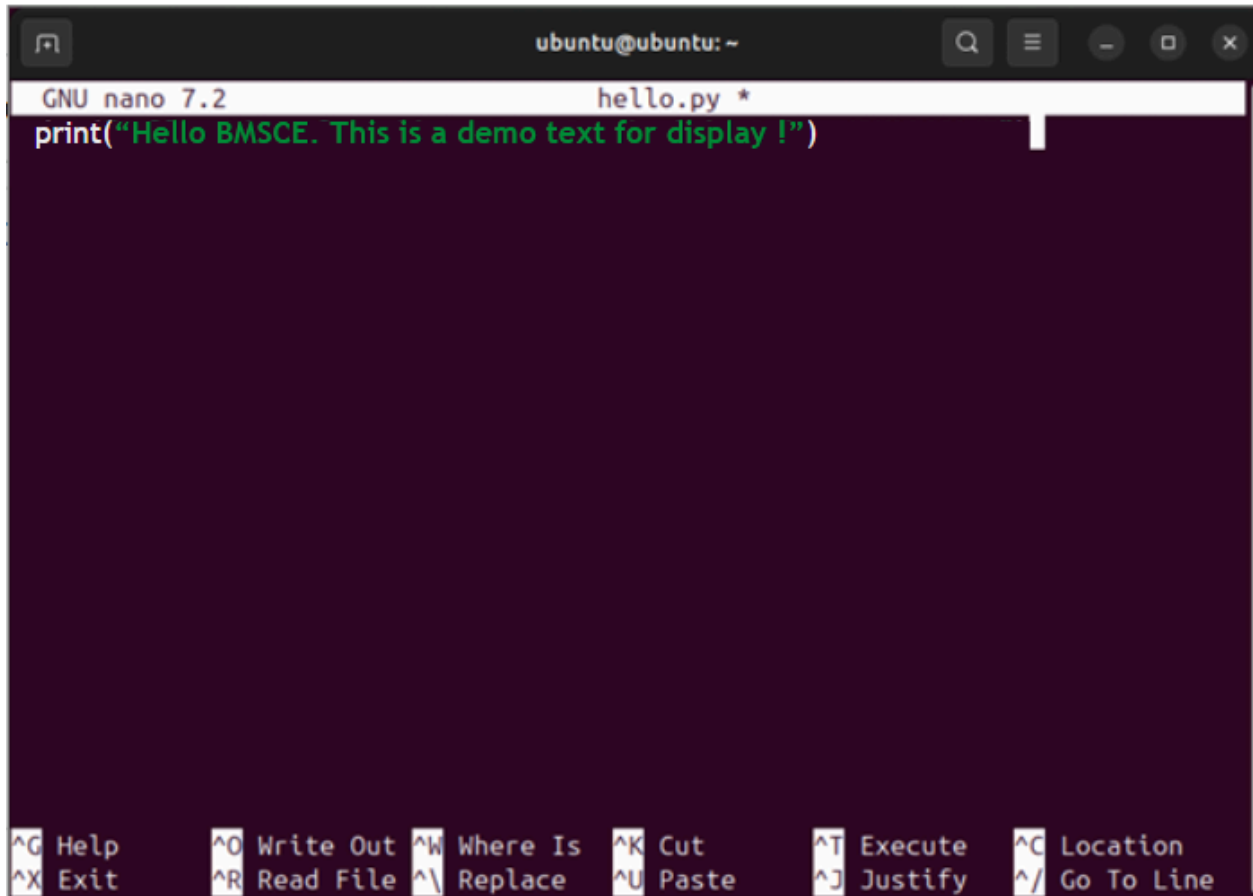
Fig 1.5 Updating package manager

```
32.3 kB]  
Get:57 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Met  
adata [1480 B]  
Get:58 http://archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Compone  
nts [216 B]  
Get:59 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Package  
s [780 B]  
Get:60 http://archive.ubuntu.com/ubuntu noble-backports/multiverse i386 Packages  
[780 B]  
Get:61 http://archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-e  
n [372 B]  
Get:62 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Compone  
nts [212 B]  
Fetched 19.2 MB in 60s (322 kB/s)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
185 packages can be upgraded. Run 'apt list --upgradable' to see them.  
ubuntu@ubuntu:~$ sudo apt install python3 python3-pip -y  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
python3 is already the newest version (3.12.3-0ubuntu2.1).  
python3 set to manually installed.
```

Fig 1.6 Installing python

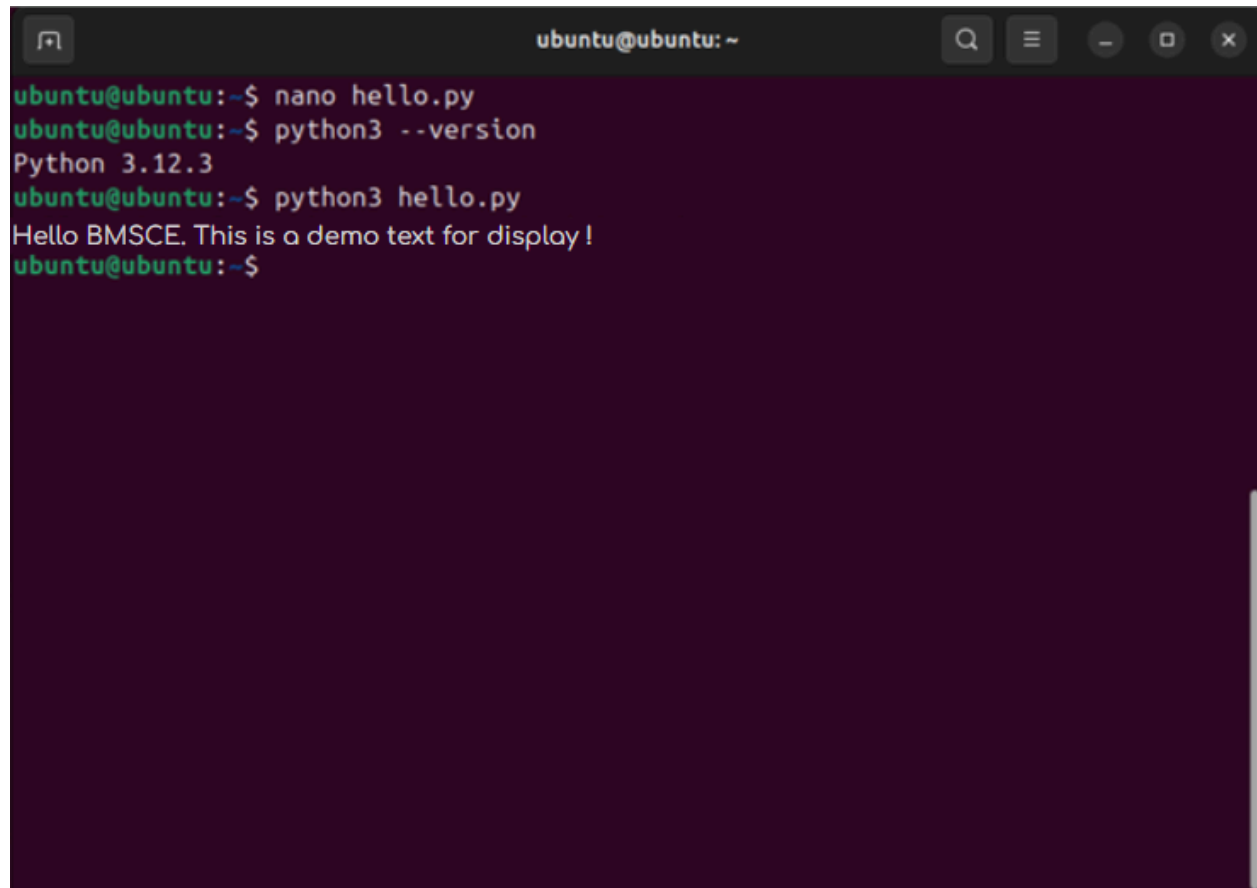
```
ubuntu@ubuntu: ~  
Setting up dpkg-dev (1.22.6ubuntu6.5) ...  
Setting up gcc-13 (13.3.0-6ubuntu2~24.04.1) ...  
Setting up python3.12-dev (3.12.3-1ubuntu0.12) ...  
Setting up libpython3-dev:amd64 (3.12.3-0ubuntu2.1) ...  
Setting up g++-13-x86-64-linux-gnu (13.3.0-6ubuntu2~24.04.1) ...  
Setting up gcc-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...  
Setting up gcc (4:13.2.0-7ubuntu1) ...  
Setting up python3-dev (3.12.3-0ubuntu2.1) ...  
Setting up g++-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...  
Setting up g++-13 (13.3.0-6ubuntu2~24.04.1) ...  
Setting up g++ (4:13.2.0-7ubuntu1) ...  
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode  
Setting up build-essential (12.10ubuntu1) ...  
Processing triggers for gnome-menus (3.36.0-1.1ubuntu3) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...  
Processing triggers for systemd (255.4-1ubuntu8.12) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
Processing triggers for desktop-file-utils (0.27-2build1) ...  
Killed  
ubuntu@ubuntu:~$ python3 --version  
Python 3.12.3  
ubuntu@ubuntu:~$  
Progress: [ 99%] [#####.]
```

Fig 1.7 Installation successful



```
ubuntu@ubuntu: ~  
GNU nano 7.2 hello.py *  
print("Hello BMSCE. This is a demo text for display !")  
  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```

Fig 1.8 Python code

A terminal window with a dark background and light-colored text. The window title bar shows 'ubuntu@ubuntu: ~' and standard window controls. The terminal displays a sequence of commands and their outputs: 'nano hello.py' (no output), 'python3 --version' (output: 'Python 3.12.3'), and 'python3 hello.py' (output: 'Hello BMSCE. This is a demo text for display!'). The prompt 'ubuntu@ubuntu:~\$' is visible at the end of each line.

```
ubuntu@ubuntu:~$ nano hello.py
ubuntu@ubuntu:~$ python3 --version
Python 3.12.3
ubuntu@ubuntu:~$ python3 hello.py
Hello BMSCE. This is a demo text for display!
ubuntu@ubuntu:~$
```

Fig 1.9 Final python program output